Jialing Wu

The Ohio State University,
Department of Engineering Education

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Education

The Ohio State University, Department of Engineering Education — Expected May 2028
Ph.D. in Engineering Education
Minor in Computer Science and Engineering — AI Track
Advisor: Dr. Adam Carberry

Vanderbilt University, Peabody College — May 2024
Master of Education, International Education Policy and Management

Shanghai University — July 2017
Bachelor of Engineering, Mechanical Engineering

Research Interests

Methodological Core: AI and computational methods (machine learning, LLMs, text analysis, network analysis, GIS, person-centered methods)
Application Areas: Educational research, teaching practice, and student learning; engineering education; equity and access

Publications

Peer-Reviewed Journal Articles

[22] **Wu, J.**, Shehab, F., & McGuire, L. (in press). Too deep or too shallow: Reflections on methods training from the perspective of engineering education students. *Teaching Educational Research Methods*.

[21] Yan, J., Wei, X., & **Wu, J.** (2026). Curriculum- and grade-level AI policy implementation: A comparative analysis of North Carolina and California. *Educational Policy*. https://doi.org/10.1177/08959048261481543

[20] **Wu, J.**, Dalal, M., Figard, R., & Carberry, A. R. (2026). Understanding transgender and gender-nonconforming students' experiences in precollege engineering education: A latent profile analysis [Invited contribution based on ASEE PCEE Division Best Paper award]. *Journal of Pre-College Engineering Education Research (J-PEER)*, *16*(1), Article 3. https://doi.org/10.7771/2157-9288.1515

Policy Insights and Editorials

[19] Darling-Aduana, J. S., Heinrich, C., Noonan, J., **Wu, J.**, & Enriquez, K. (2026). Failing to learn from failure in online credit recovery assessments [Policy Insights]. *Journal of Policy Analysis and Management*, *45*(3), e70120. https://doi.org/10.1002/pam.70120

Book Chapters

[18] Yan, J., **Wu, J.**, & Wei, X. (2026). Silent policies: How GenAI guidelines in U.S. higher education overlook international students' experiences. In A. Chen (Ed.), *Generative AI for supporting international student well-being in higher education* (pp. 95–132). IGI Global. https://doi.org/10.4018/979-8-3373-6571-8.ch004

Conference Papers

[17] **Wu, J.**, Ren, S., & Wang, S. (2026). Using bipartite network analysis to examine global collaboration between engineering and STEM education (2014–2024). *2026 ASEE Annual Conference & Exposition Proceedings*, Charlotte, NC. https://doi.org/10.18260/1-2--61184

[16] **Wu, J.**, & Wang, S. (2026). What does the "person-centered quantitative approach" mean in practice? A literature review in STEM education research [Work-in-progress paper]. *2026 ASEE Annual Conference & Exposition Proceedings*, Charlotte, NC. https://doi.org/10.18260/1-2--60763

[15] **Wu, J.**, & Carberry, A. R. (2026). Integrating robotics and engineering education: Initial validation testing of an SCCT- and CT-based survey instrument [Work-in-progress paper]. *2026 ASEE Annual Conference & Exposition Proceedings*, Charlotte, NC. https://doi.org/10.18260/1-2--60118

[14] **Wu, J.**, & Melo de Lyra, M. V. (2026). Exploring perspectives on pre-college robotics education: A social media–based topic and emotion analysis [Work-in-progress paper]. *2026 ASEE Annual Conference & Exposition Proceedings*, Charlotte, NC. https://doi.org/10.18260/1-2--59786

[13] Melo de Lyra, M. V., **Wu, J.**, & Carberry, A. R. (2025). Exploring early-career engineering faculty experiences teaching using social cognitive theory. *2025 IEEE Frontiers in Education Conference (FIE)*, Nashville, TN, pp. 1–10. https://doi.org/10.1109/FIE63693.2025.11328485

[12] Dalal, M., **Wu, J.**, & Iqbal, A. (2025). Overcoming challenges in robotics education: Examining teacher facilitators and barriers. *Research in Engineering Education Symposium (REES)*, Johor Bahru, Malaysia.

[11] **Wu, J.**, Dalal, M., & Carberry, A. (2025). Advancing equity: Exploring the experiences of transgender and gender non-conforming students in a pre-college engineering course [Work-in-progress paper]. *2025 ASEE Annual Conference & Exposition Proceedings*, Montreal, QC, Canada. https://doi.org/10.18260/1-2--57577

[10] Wei, X., **Wu, J.**, & Klein-Gardner, S. S. (2025). Parents' knowledge, attitude, and behavior on pre-college engineering education course [Work-in-progress paper]. *2025 ASEE Annual Conference & Exposition Proceedings*, Montreal, QC, Canada. https://doi.org/10.18260/1-2--55568

[9] Lachapelle, C. P., Dalal, M., McKeown, K., & **Wu, J.** (2025). Curriculum design for all learners. *2025 ASEE Annual Conference & Exposition Proceedings*, Montreal, QC, Canada. https://doi.org/10.18260/1-2--56190

[8] **Wu, J.**, & Dalal, M. (2024). High school students' perspectives on pre-college engineering education courses (Fundamental). *2024 ASEE Annual Conference & Exposition Proceedings*, Portland, OR. https://doi.org/10.18260/1-2--47527

[7] **Wu, J.**, Léger, N., & Klein-Gardner, S. (2024). High school students' perspectives on mathematical modeling in the engineering design process (RTP). *2024 ASEE Annual Conference & Exposition Proceedings*, Portland, OR. https://doi.org/10.18260/1-2--47528

Conference Presentations

[6] **Wu, J.**, & Li, D. (2026, November 5–7). Teachers as AI policy actors: Focus group evidence from a K–12 robotics consortium [Poster presentation]. *Association for Public Policy Analysis & Management (APPAM) Annual Fall Research Conference*, Boston, MA.

[5] Wang, J., **Wu, J.**, & Cheng, Y. (2026, July 28–31). Cross-cultural evaluation of LLMs: A mixed-method study of Chinese-to-English movie title translation [Poster presentation]. *12th International Conference for Computational Social Science (IC2S2 2026)*, Burlington, VT.

[4] Wei, X., & **Wu, J.** (2026, April 8–12). Navigating AI in higher education: Student views on professors' attitudes toward AI tools in the classroom analyzed through sentiment analysis and topic modeling [Conference presentation]. *American Educational Research Association (AERA) Annual Meeting*, Los Angeles, CA.

[3] **Wu, J.**, & Yan, J. (2026, March 28 – April 1). Comparing high school accessibility in California and New York: A GIS-based assessment [Conference presentation]. *Comparative and International Education Society (CIES) Annual Conference*, San Francisco, CA.

[2] Wei, X., **Wu, J.**, Klein-Gardner, S., & Dalal, M. (2025, November 10–14). Evaluating fidelity of teaching in pre-college engineering: A case study of curriculum implementation [Poster presentation]. *Evaluation 2025*, Kansas City, MO. eval25.eventscribe.net

[1] Darling-Aduana, J. S., Heinrich, C., Noonan, J., **Wu, J.**, & Enriquez, K. (2025, April 23–27). Failing to learn from failure: The facade of online credit recovery assessments [Conference presentation]. *American Educational Research Association (AERA) Annual Meeting*, Denver, CO. https://doi.org/10.3102/ip.25.2182630

Manuscripts Under Review / In Preparation

[2] **Wu, J.**, Wei, X., & Yan, J. (under review). Engineering education deserts in the United States: A geographic random forest analysis of broad-access engineering programs in higher education. *AERA Open*.

[1] Berhane, B., Dalal, M., Ross, L., **Wu, J.**, & Wang, J. (under review). Examining student- and school-level outcomes in pre- and post-survey data from a pre-college engineering program. *Journal of Engineering Education*.

Grants, Fellowships, and Awards

Grants

Institute for Humane Studies Grant Proposal ($7,500 requested; pending) — *Higher Education Technology Instructors' Professional Identity in the Age of Generative AI* 2026

Smith Travel Grant ($1,000; awarded), Department of Engineering Education, OSU 2025, 2026

Career Development Grant ($350; awarded), Council of Graduate Students, OSU 2025

Fellowships and Scholarships

Frédéric Bastiat Fellowship ($5,000), Mercatus Center at George Mason University 2025–2026

Scholastic Achievement Scholarship ($26,500), Vanderbilt University, Peabody College 2022–2024

University Scholarship for Academic Excellence (¥1,000), Shanghai University 2014, 2015

Honors and Awards

Best Paper Award, 2nd Place, Pre-College Engineering Education Division, ASEE — transgender and gender-nonconforming students in pre-college engineering 2025

Professional Experience

Research Experience

The Ohio State University, Columbus, OH August 2024 – Present
Graduate Research Associate, Department of Engineering Education
(with Dr. Adam Carberry)

The e4usa+FIRST program pairs the e4usa curriculum with FIRST Robotics for high school students.

- Lead the development of survey instruments grounded in Social Cognitive Career Theory (SCCT) and computational thinking to assess student self-efficacy, identity, and skill development.
- Collect pre-test data via Google Forms for initial validity checks of the survey instrument.
- Prepare and submit IRB applications and protocols across multiple high school program sites.
- Conducted qualitative analysis of teacher interviews on robotics curriculum implementation, identifying facilitators and barriers; co-authored methods and results sections.

Vanderbilt University, Nashville, TN June 2023 – July 2024
Graduate Research Assistant, School of Engineering
(with Dr. Stacy Klein-Gardner, Dr. Medha Dalal)
e4usa is a national initiative introducing engineering design principles to high school students.

- Investigated high school students' perspectives on engineering courses through qualitative analysis of post-test responses; developed codebooks and conducted pre/post statistical comparisons across cohorts.
- Analyzed focus group data on mathematical modeling tools through cognitive load theory.
- Co-authored two ASEE 2024 conference papers; led research group meetings and managed timelines.
- Revised Unit 2 curriculum (water filters → wind turbines), redesigning lessons and MATLAB code.
- Drafted facilitator memos for summer teacher PD programs and managed Canvas course materials.

Teaching Experience

The Ohio State University, Columbus, OH August 2026 – May 2027
Graduate Teaching Associate, Department of Engineering Education
ENGR 1181: Fundamentals of Engineering I

- Facilitate four weekly laboratory sections for first-year engineering students, delivering short instructional presentations and guiding 80-minute hands-on engineering activities.
- Coordinate with faculty instructors and undergraduate teaching assistants to implement labs, manage attendance and team processes, and apply course policies consistently across sections.
- Assess laboratory assignments and support students through office hours, make-up arrangements, individualized feedback, and accessible instructional practices.

Industry Experience

Howard Hughes Medical Institute (HHMI), Chevy Chase, MD May 2024 – August 2024
Assessment Intern, Science Education (BioInteractive Team)

- Developed an HTML codebook (9 chapters, 25 sections) guiding accessible Canvas course design.
- Conducted UDOIT accessibility audits and authored a guideline handbook for content editors.

Shanton Way Executive Search, Shanghai, China July 2017 – July 2022
Consultant

- Executive recruitment for clients in international K–12 education, consulting, finance, and technology sectors.
- Sourced and managed a candidate pool of 1,500+ bilingual professionals across international schools.

Workshops and Facilitation

From Reddit to Policy Insights: Using AI to Summarize Public Opinion on Robotics Courses November 7, 2026
Facilitator, AI Peer Learning Lab, "From Text to Evidence: AI-Assisted Analysis and Measurement," 2026 APPAM Fall Research Conference, Boston, MA

Training and Certificates

College and University Teaching Graduate Certificate In Progress
The Ohio State University — Selected coursework: College Teaching; Learning Theory, Pedagogy, and Assessment in Engineering Education

Summer Institute in Computational Social Science 2025
University of Rochester — Topics: LLMs, network analysis

ICPSR Summer Program 2024
University of Michigan — Topics: Python programming, R programming, text analysis

Service and Leadership

Professional Service

Reviewer; Journal of Engineering Education	2026
Reviewer; AERA Annual Meeting	2026
Reviewer; ASEE Annual Conference & Exposition	2025, 2026
Discussant; APPAM Annual Fall Research Conference	2026
Student Session Moderator; ASEE Annual Conference & Exposition	2026
Reviewer; SACNAS NDiSTEM Conference	2025

Community Service

Institutional Coordinator; Engineering Education Graduate Student Exchange Program, OSU	2025–Present
Graduate Ambassador; Department of Engineering Education, OSU	2025–Present
Graduate Student Host; Faculty Candidate Visits, Department of Engineering Education, OSU	2024–2026
Volunteer; Florence Nightingale Day for Statistics and Data Science, OSU	2025
Founding Member and Treasurer; iMAP, Vanderbilt University	2023–2024
Mentor; Peabody Peer Career Mentor Program, Vanderbilt University	2023–2024
Mentor; Peabody International Students and Affairs, Vanderbilt University	2023–2024
Migrant Children Program Coordinator and Teacher; AIQUZHI Volunteer Group	2017–2022

Professional References

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